

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0649

Loudon 345 kV
May 26, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0649. The Interconnection Customer proposes a 414.7 MW natural gas combined cycle facility interconnecting at the Loudon 345 kV substation in the Virginia area. The request was received on July 19, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 232 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,628,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Indian Lake – Albert Pike Substation 138.0 kV	Loudon Substation – Thomaston 345.0 kV	120.5	307	45.8
Brambleton – Yardley Ridge 138.0 kV	Loudon – Thomaston 345.0 kV	112.6	372	45.8
Brambleton – Albert Pike 138.0 kV	Loudon – Thomaston 345.0 kV line	107.5	358	45.8
Lodon – Yardley Ridge 138.0 kV Line	Loudon – Thomaston 345.0 kV	105.5	534	64.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Albert Pike Substation – Indianlake 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,036,000	100% (\$4,036,000)
Brambleton – Yardley Ridge 138.0 kV	Rebuild line with higher-capacity conductor	\$2,839,000	100% (\$2,839,000)
Albert Pike – Brambleton Substation 138.0 kV	Reconductor line	\$1,031,000	100% (\$1,031,000)
Yardley Ridge – Loudon 138.0 kV Line	Reconductor line	\$722,000	100% (\$722,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$8,628,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.